

The Law of Fairness: Terminal neutrality as a boundary condition on conscious state space

Abstract

Summary: The Law of Fairness hypothesizes terminal neutrality as a latent structural boundary condition on unified conscious trajectories, not as karma but as a testable candidate physical constraint on conscious state space. Phenomenologically, the model represents the life ledger as $L(t) = \int_0^t g_\theta(s) F(s) ds$; in its strong latent form it posits $L(T) = 0$ as a structural idealization within a reserve-coupled architecture. Empirically, however, the model does not claim direct observation of exact equality; it predicts preterminal variance compression, terminal concentration near neutrality, and equivalence-band neutrality in physiological ledger estimators. Within the formal stochastic realization, this closure is made dynamically achievable by the multiplicatively coupled process $L(t) = U(t)Y(t)$ in which declining biological reserve $U(t)$ compresses fluctuations in the ledger by scaling the effective diffusion coefficient $\sigma U(t)$. Because the endpoint T is defined through a separately preregistered biological signal that is not algebraically reducible to the ledger itself, the prediction avoids tautology and remains empirically falsifiable. The framework therefore stands or falls on observable signatures in physiological telemetry and, where available, longitudinal affect data, evaluated under preregistered observation models and compared against preregistered simpler baseline models. It is presented as a formal hypothesis for the academic community to test and attempt to falsify.

The Law of Fairness (LoF) is presented as a formal, falsifiable framework. It invites rigorous empirical scrutiny and cross-disciplinary validation according to the standard methodologies employed in biophysics, systems neuroscience, and mathematical biology. The validity of the LoF depends upon its ability to withstand robust falsification protocols. If it is true, it leaves constraint signatures not reproduced by ordinary unconstrained homeostasis, hedonic adaptation, or ensemble-based reinforcement learning without added state-coupled terminal control. The framework therefore treats fairness not as a moral ideal but as a candidate physical constraint on the trajectory of conscious state space. Each proposed mechanism in this framework is motivated by published findings across affect dynamics, sleep physiology, allostatic energetics, horizon-dependent valuation, and inhibitory control. The theoretical scaffolding is explicit, the empirical alignments are stated, and the falsification criteria are explicit. Ultimately, the framework's utility will be determined by prospective data and out-of-sample predictive accuracy.

Terminology and Core Parameters

To eliminate semantic ambiguity, we define the parameters strictly:

- $\mathbf{F}(t)$: latent valence-rate signal in the phenomenological interpretation of the model.
- $\mathbf{z}_k(t)$: preregistered intensive, non-conservative physiological contribution features transformed into signed channels relative to preregistered anchors or subject-specific baselines defined by preregistered rules, so that the sign of $\mathbf{z}_k(t)$ carries burden-increasing versus restorative contribution under a preregistered sign convention.
- $\mathbf{HCI}(t)$: Hedonic Composite Index; preregistered empirical estimator built from signed contribution channels $\mathbf{z}_k(t)$ with nonnegative weights w_k .
- $\mathbf{L}(t)$: latent cumulative ledger. Phenomenologically, $L(t) = \int_0^t g_\theta(s) F(s) ds$ denotes the unity-gated valence-integral idealization, with g_θ made explicit later; in the stochastic formalization, the ledger is modeled by the reserve-coupled process $L(t) = U(t)Y(t)$. These are phenomenological and stochastic representations of the same underlying construct rather than two simultaneous primitive definitions.
- $\mathbf{Y}(t)$: core normalized stochastic process (semimartingale) representing the underlying valence dynamic independent of biological reserve scaling.
- $\hat{\mathbf{L}}(T) = \sum \mathbf{HCI}(t_i) \Delta t_i$: measured ledger estimator.
- Δt_i : discrete sampling interval for the empirical estimator.
- $\theta(t)$: Unity Index (proxy for conscious access unity operationally separated from the HCI construction, e.g., perturbational complexity indices; Casali et al., 2013).
- $\mathcal{Y}_t$: observation filtration; the history of accessible information up to time t .
- T : endpoint stopping time (Unity Index threshold crossing), defined with respect to the observation filtration $\mathcal{Y}_t$; operationally, the stopping event itself is determined only by preregistered unity-proxy channels rather than by the ledger estimator.
- $\mathbf{U}(t)$: latent biological reserve / plasticity process, operationalized empirically through preregistered reserve proxies.
- $\mathbf{H}(t)$: filtered conditional estimate of the remaining horizon, defined with respect to the controller's observation history rather than inaccessible full latent information; central control variable governing urgency, admissibility, and compensability pressure.
- $\Psi(t)$: Viability Ratio defined as $(\Psi(t) = |L(t)| / U(t))$; a dimensionless, reserve-normalized latent-amplitude metric summarizing imbalance relative to available biological reserve. Under the core multiplicative representation $L(t) = U(t)Y(t)$, $\Psi(t) = |Y(t)|$ when $U(t) > 0$.
- Φ : compensability score / future-preserving admissibility weight.
- $\lambda(t)$: shadow price / value-gradient penalty weighting compensability as horizon collapses.

Introduction and Core Hypothesis

The Law hypothesizes terminal neutrality as a latent structural boundary condition on unified conscious trajectories. In its strongest latent reading, it posits exact pathwise closure rather than an ensemble tendency:

$$P(L(T) = 0) = 1$$

in the underlying idealized process. Empirically, however, the public scientific claim is not direct observation of exact equality, but the measurable consequence structure generated by that latent architecture, namely preterminal variance compression, terminal concentration near zero, and equivalence-band neutrality of $\tilde{L}(T)$ under explicit proxy uncertainty.

A unified conscious life is modeled as a single, time-irreversible realized trajectory of a non-ergodic process terminating at a stopping time defined by the Unity Index threshold and linked to reserve collapse through the preregistered $\theta(t)$ and $U(t)$ relationship rather than by the ledger itself. The strongest public empirical targets are therefore preterminal variance compression, terminal concentration near zero, and equivalence-band neutrality rather than direct observation of exact equality.

Multiplicative Coupling and Itô Dynamics. To avoid mathematical tautology, the ledger is multiplicatively coupled to the biological reserve $U(t)$, representing residual epigenetic and metabolic plasticity. $U(t)$ is modeled on the preterminal interval as a finite-variation reserve process that remains strictly positive for $t < T$ and decays toward zero as the terminal regime is approached:

$$dU(t) = -v(t) dt$$

Let $Y(t)$ be an unconstrained diffusion process defined by:

$$dY(t) = \sigma dW(t)$$

with initial state $Y(0) = Y_0$. The coupled ledger is defined by the product representation:

$$L(t) = U(t)Y(t)$$

Applying Itô's Lemma yields the governing dynamics:

$$dL(t) = - \left(\frac{v(t)}{U(t)} \right) L(t) dt + \sigma U(t) dW(t)$$

As $U(t) \rightarrow 0$ near the endpoint, two critical empirical signatures emerge:

- **Drift Dominance:** Under the maintained assumption that there exists $\varepsilon > 0$ such that $v(t) \geq \varepsilon$ for all t sufficiently close to T , the mean-reversion rate $v(t)/U(t)$ diverges as $U(t) \rightarrow 0$, generating increasingly strong mean-reversion pressure toward zero.

- **Variance Compression:** The diffusion coefficient $\sigma U(t)$ vanishes as $U(t)$ approaches zero, suppressing stochastic excursions and producing increasing concentration of the probability mass of $L(t)$ near zero.

These dynamics, together with the shrinking filtered horizon $H(t)$, support steep inverse-horizon weighting and predict aggressive pruning of high-variance trajectories via the Queue System (QS).

I. The Endpoint Firewall & Statistical Rigor

Empirical validation must prioritize the assessment of the terminal boundary condition. "Death of Mind" is defined operationally as a causal stopping time driven by a preregistered Unity Index threshold, not by somatic death. Formally:

$$T = \inf_{t \geq 0} \{ t: \theta(t) \leq \theta^0 \}$$

with the event $\{T \leq t\}$ measurable with respect to the observation filtration $\mathcal{Y}_t$. If you define "death" as "the time the ledger hits zero," then neutrality is a tautology. The LoF framework explicitly precludes this definition to prevent definitional circularity and to preserve nonidentity between the stopping rule and the ledger estimator. The Unity Index $\theta(t)$ must be derived from physiological channels that are not algebraically or instrumentally identical to those defining HCI, and any residual statistical dependence must be modeled explicitly to avoid circularity.

Because physiological telemetry must avoid exact conservative state variables, the empirical ledger is constructed from path-dependent regulatory-burden or regulatory-activity proxies rather than from quantities that collapse to $S(T) - S(0)$, preventing the path integral from reducing to a trivial boundary term. To prevent algebraic collapse, LoF mandates that empirical observables must be non-conservative, path-dependent physiological regulatory-burden or regulatory-activity proxies (e.g., allostatic-load-related regulatory-cost proxies under the Energetic Model of Allostatic Load; Bobba-Alves et al., 2022). Terminal neutrality is hypothesized as a dynamical outcome of the system's evolution rather than an a priori algebraic identity.

II. Empirical Domains & Falsification Protocols

Here are the core predictions and auxiliary mechanism-level hypotheses intended to distinguish LoF from unconstrained or simpler baseline models:

- Terminal concentration near neutrality under a separately defined stopping rule that is not algebraically reducible to the ledger estimator, together with a reserve-coupled terminal architecture.
- Variance compression scaling with preregistered reserve proxies for the latent collapse process $U(t)$.
- A specific horizon-sensitive compensability weighting predicting inhibitory-braking signatures in the brain.

- An auxiliary sleep-dependent rebalancing hypothesis that, if present, could provide one candidate offline compensatory channel.

In-Silico Falsification: The Virtual Terminal Maze. The task is formalized as an adversarial finite-horizon sequential decision problem in which agents under declining reserve $U(t)$ and shrinking horizon $H(t)$ must identify a low-arousal compensable path among high-arousal decoy options. Under LoF-style admissibility gating, the feasible policy set collapses as $H(t)$ shrinks, forcing the policy toward compensable trajectories. For intuition, consider a computer-simulated agent with severe allostatic debt placed in a virtual maze with 100 exits, 99 of which are high-arousal decoys that end in failure and one of which is compensable. Under unconstrained or myopic reward-optimizing baselines, lure-following failure rates are predicted to exceed those of controllers equipped with horizon-scaled admissibility constraints. Under the LoF architecture, as $H(t)$ shrinks and $U(t)$ approaches zero, the shadow price of compensability $\lambda(t)$ rises, pruning actions that jeopardize closure feasibility. The empirical claim is comparative rather than absolute: success rates should exceed preregistered unconstrained baselines if the horizon-scaled compensability constraint is active.

Domain 1: The Queue System & Admissible-Set Pruning. In cognitive labs, horizon-scaled $\Phi \times H(t)^{-1}$ is predicted to explain additional variance in valuation and control hubs beyond standard predictors such as utility, conflict, and arousal. Anchored in the Expected Value of Control framework (Shenhav et al., 2013), the Queue System is best interpreted normatively as a safety-critical output-feedback policy. The right inferior frontal gyrus (rIFG) and dACC are treated here as candidate neural correlates of observer-based constraint enforcement on filtered estimates of latent state rather than as a mechanically established one-to-one implementation of formal Control Barrier Functions (Ames et al., 2017; Wang & Xu, 2022). These candidate control hubs are hypothesized to brake low-compensability choices that threaten to push the estimated trajectory outside the viability manifold while explicitly tolerating physiological measurement noise and state-estimation error. Admissible menu counts are predicted to decrease approximately with $H(t)^{-1}$ and to exhibit overdispersion, tested via preregistered Negative Binomial generalized linear mixed models. If preregistered perturbation studies with adequate target engagement fail to produce the predicted directional increase in admissible-set leakage after accounting for observation noise and estimator uncertainty, the mechanism is weakened; repeated null results under adequate power and target engagement would falsify this mechanism-level claim.

Domain 2: Systems Biology & Regulatory / Energetic Cost. Unresolved negative valence is hypothesized to impose increased physiological regulatory cost. High-variance trajectories are hypothesized to correlate with increased regulatory burden and, under allostatic-load frameworks, with acceleration in biological aging markers (Juster et al., 2010; Bobba-Alves et al., 2022), providing a plausible physiological correlate of $U(t)$ decay. If preregistered latent imbalance estimates or validated multimodal burden measures drift into persistent deficit without

corresponding acceleration in preregistered reserve proxies for $U(t)$, the proposed biological anchoring is weakened.

Domain 3: Horizon Scaling & Neural Revaluation. If this mechanism is correct, valuation systems centered on vmPFC are predicted to encode a distinct value surplus for highly compensable, reparative choices. The preregistered neural target is a positive $\Phi \times H(t)^{-1}$ interaction in BOLD/EEG signals after adjustment for standard value, conflict, and arousal covariates.

Domain 4: Sleep Physiology & Noradrenergic Blockade. As an auxiliary mechanism-level hypothesis rather than a load-bearing core prediction, the model permits the possibility that when waking life offers no behavioral path to balance, a subset of healthy REM processes contributes to a compensatory shift toward more neutral or mastery-themed states (extending Cartwright et al., 1998). Mechanism: reduced noradrenergic tone during healthy REM is hypothesized to permit affective reweighting with attenuated autonomic carryover. Caveat & Falsifier: REM's noradrenergic suppression is documented to fail in PTSD-like physiology (Germain et al., 2008). This is a quantifiable boundary for the proposed REM mechanism: if recurrent pathological failures prevent this rebalancing pattern at a preregistered cohort-level prevalence exceeding a bound justified from pilot or cohort-specific data, the REM-channel hypothesis is rejected and the broader LoF framework must rely on alternative compensatory pathways. While hypothesized as a modifiable vulnerability factor, bidirectional associations between PTSD and sleep disturbances are acknowledged; preregistered longitudinal designs should assess temporal asymmetry and lagged directional structure using cross-lagged or closely related longitudinal models.

Domain 5: Social Coupling & Scarcity. This is an exploratory extension rather than a core falsifier. The framework predicts an emergent shadow price on scarce relief opportunities, prioritizing those nearer closure. Failure of this prediction does not invalidate the core LoF boundary condition but would remove this auxiliary mechanism.

Domain 6: Gerontology & Terminal Variance Compression. If reserve proxies associated with $U(t)$ are entering a low-reserve terminal regime, physiological flexibility measures (e.g., HRV) are expected on average to show reduced variability or complexity, and reserve-stratified cross-sectional ledger distributions, together with within-trajectory rolling ledger variance, are predicted to contract. Under standard TOST, neutrality is supported only if both one-sided tests reject the null, meaning the 90% confidence interval for the measured estimator $\hat{L}(T)$ lies entirely within $[-K, +K]$. To prevent subjective tuning, TOST is supplemented with Bayes factors computed under a preregistered weakly informative prior family together with preregistered sensitivity analysis over reasonable prior scales. $BF_{01} > 30$ (strong evidence) favors the neutrality model over the imbalance alternative, and $BF_{10} > 30$ favoring terminal imbalance constitutes a preregistered dataset-level falsifier that contributes to theory-level falsification under the cohort logic specified later.

III. Implementation and Future Research Directions

Preregistration packages, HCI code templates, power-analysis scripts, and ethical templates are being prepared. Adversarial replications, alternative model fits, and null results are explicitly welcomed under the framework described here and its intended preregistered implementation.

Quickstart Falsification Tests (Using Standard Existing Modalities):

- **Terminal Variance Compression (Hospice):** Fit preregistered longitudinal models of HCI-based or other preregistered ledger-proxy variance, and where available complementary affect-variance measures, versus time-to- T and reserve or unity proxies, using a unity-proxy-defined T or its preregistered state-space estimate; where feasible, use joint longitudinal-endpoint models so that reserve telemetry, informative missingness, left-truncation or survivor-selection effects, and the stopping process are estimated coherently. Preregister that variance is predicted to contract as the terminal regime is approached, with major medication, sedation, missingness, and survivor-selection confounds modeled explicitly.
- **Horizon $\times$ Compensability (Decision Tasks):** Preregister a $\Phi \times H(t)^{-1}$ interaction predicting choice signals.
- **REM Rebalancing Channel (Sleep Labs):** Test whether high negative waking load L_{wake} (waking ledger accumulation prior to sleep) predicts within-night REM affective rebalancing across successive REM periods in the subsequent sleep episode.

Model Parsimony and Comparative Testing: If a parsimonious adversarial model, whether based on standard homeostatic regulation, endpoint-conditioned Brownian-bridge dynamics, positive-terminal-affect drift baselines, ordinary memory consolidation, age-only decline, frailty-only decline, medication-burden-only dynamics, or modern constrained-control / safe-RL baselines, yields equivalent or superior held-out prediction of terminal behavior, reserve-stratified variance compression, terminal concentration near neutrality, and horizon effects without reserve-coupled terminal structure, the LoF framework should be rejected in favor of the simpler alternative. Preregistered comparison should rely on proper out-of-sample predictive criteria and calibration diagnostics rather than in-sample fit alone.

The question is no longer philosophical; it is strictly empirical. The scientific utility of this hypothesis rests not on its theoretical novelty, but on its exposure to rigorous empirical falsification.

Law of Fairness: Mathematical Formalization and Stochastic Dynamics

The Law of Fairness (LoF) posits a hypothesized boundary condition that, in its strongest latent reading, is universal across unified conscious streams and remains a testable conjecture awaiting empirical confirmation across diverse populations. Let $F(t)$ denote the latent valence-rate signal of unified conscious experience. In the phenomenological representation, the life ledger is written as the unity-gated time integral:

$$L(t) = \int_0^t g_\theta(s) F(s) ds$$

where $g_\theta(t)$ ensures that accrual occurs only during periods of unified conscious access. In the strong latent form of the Law of Fairness, the terminal condition is:

$$L(T) = 0$$

The ungated expression $\int_0^T F(t) dt = 0$ should be read only as a shorthand conceptual statement of terminal neutrality not as the operational phenomenological definition used in the formal model. While hypothesized as universal, initial empirical tests will focus on human populations, with extensions to non-human conscious systems (e.g., via animal models) requiring separate validation.

In the stochastic formalization used below, that same underlying construct is represented by the reserve-coupled semimartingale $L(t) = U(t)Y(t)$. This representation should be read as a model-level realization of the same candidate ledger concept rather than as a claim that the phenomenological rate $F(t)$ and the stochastic process $U(t)Y(t)$ are pointwise identical, and not as two simultaneously primitive definitions.

Strong Form LoF (latent structural idealization): the strongest latent reading is that every unified conscious stream satisfies exact terminal closure in the underlying reserve-coupled idealization. Empirically, however, the model is tested only through preregistered measurable corollaries, namely variance compression, terminal concentration near zero, and equivalence-band neutrality of $\dot{L}(T)$ under explicit observation error. In this scientific sense, the public claim is not direct observation of exact equality but successful recovery of its measurable consequence structure.

The Shadow Price of Admissibility: terminal neutrality functions as a binding boundary condition on admissible trajectories in conscious state space. In control-theoretic terms, residual imbalance at collapse corresponds to terminal feasibility violation, and trajectories violating the boundary are excluded by the model's imposed constraint. Active-inference or predictive-processing interpretations may be layered onto this picture (e.g., Hesp et al., 2021; Smith et al., 2021), but they are not required for the formal model itself.

Epistemological Framework (Testing): because $L(t)$, $U(t)$, and the controller-relevant horizon are latent quantities, the model must be interpreted as a partially observed stochastic control system (e.g., Kara & Yüksel, 2024). Both the organism and the researcher operate under incomplete information; the organism acts under partial observation, and the researcher infers latent state through physiological telemetry. Empirical tests therefore rely on preregistered observation models that map measurable signals into probabilistic estimates of latent imbalance, reserve, and stopping-time distribution rather than assuming direct access to the ledger itself; confirmatory support additionally requires preregistered model-recovery and out-of-sample comparison analyses showing that the reserve-coupled architecture is empirically distinguishable from simpler baselines under realistic noise. When useful, these filtered latent-state estimates may be summarized by a belief state $\pi(t)$ over the control-relevant hidden variables.

One preregistered observation channel for the latent valence-rate process is a Hedonic Composite Index (HCI) constructed from longitudinal physiological telemetry. Crucially, these physiological observables (e.g., heart rate variability, endocrine markers) are not assumed to be literal, direct energy meters (calorimetry); rather, they function as hypothesized estimators for latent regulatory burden and closure feasibility under noisy observation. While HCI provides objective telemetry, it is not a one-to-one meter of subjective valence; it functions as an observation channel within a latent-state model of imbalance and regulatory burden.

Where feasible, preregistered experience sampling methods (ESM) should be integrated as a secondary validator, and evaluation should include concordance, calibration, temporal stability, posterior-predictive adequacy, and discriminant validity against neighboring but nonidentical constructs such as generic arousal, nociception, sedation, and medication exposure, rather than reliance on a single cutoff alone. Failure to meet preregistered adequacy criteria triggers exploratory proxy recalibration requiring prospective confirmation in a new preregistered dataset, or hypothesis rejection.

Because both ESM and physiological telemetry are vulnerable to missingness, dropout, compliance bias, nuisance covariance, and construct contamination, preregistered missing-data models, negative-control channels, sensitivity analyses, and explicit nuisance-factor comparisons are required. If the proxy system cannot demonstrate adequate reliability, discriminant validity against preregistered nuisance constructs, and non-conservative path dependence, the operationalization fails and LoF receives no empirical support from that dataset. For invariance, failure of the level required for a given comparison restricts interpretation of that comparison accordingly rather than automatically invalidating all within-group analyses. Multimodal fusion (e.g., Bayesian integration of fMRI/ESM/HRV) will quantify uncertainty.

Valence polarity is treated operationally in the formal model and does not require commitment to a specific free-energy or predictive-processing ontology. Active-inference interpretations may be explored as optional overlays, but the core formal model only requires that $F(t)$ represent a latent valence-rate process linked to preregistered physiological estimators. In this sense, $F(t)$ is a

reduced-order summary of the valenced burden-regulation stream relevant to the boundary condition under study, not a claim that consciousness is exhaustively one-dimensional.

The composite weights w_k are constrained to remain nonnegative, with the sign convention carried by the preregistered channel transforms $z_k(t)$; sensitivity tests then assess asymmetric costs without permitting post hoc sign switching in the weights. When negativity bias is modeled, it is implemented through preregistered sign-asymmetric channel transforms or split-channel coding of negative and positive components, testing $L(T)$ robustness over illustrative, domain-sensitive bias ranges rather than treating any single ratio as universal; example sensitivity ranges may be informed by sources such as Baumeister et al., 2001 for general negativity and Gottman, 1994, 1999 for relationship-relevant contexts, with preregistered tests also exploring cultural moderators (e.g., individualism scores per Hofstede et al., 2010). Failure to detect meaningful moderation would limit claims about cross-cultural modulation but would not by itself falsify the core boundary-condition hypothesis.

Repeated physiological sampling and digital phenotyping treat longitudinal measurements (e.g., cortisol levels, HRV windows, wearable-derived autonomic features) as preregistered surrogate indicators of physiological regulatory cost rather than literal metabolic expenditure rates. The observation model should therefore be preregistered as a context-specific measurement model using monotone mappings, with approximately linear relationships invoked only where analytically and biologically justified and with non-linear alternatives evaluated in sensitivity analyses, including non-parametric models such as GAMs and subject-specific mixed-effects models. Proxy validity across heterogeneous populations should be evaluated through analytic-validation diagnostics, bootstrapped variance estimates, and related preregistered checks linking HRV variance and allied telemetry to physiological regulatory-cost proxies under the stated context of use, extending allostatic-load and allostatic-energetics frameworks (Juster et al., 2010; Bobba-Alves et al., 2022). Future work must calibrate these mappings against independent physiological benchmarks where feasible. This supports only provisional estimation of biological regulatory-cost proxies, not direct measurement of affective dynamics or literal metabolic expenditure (McEwen & Wingfield, 2010).

We do not identify $F(t)$ with a literal derivative of cumulative allostatic wear across the whole trajectory. Rather, unresolved burden contributes one negative component to $F(t)$, while restorative or compensatory dynamics may contribute positive components; all components are operationalized from preregistered, path-dependent, non-conservative observables rather than from gradients of a conservative state variable. If affect were modeled strictly as the derivative of a conservative state function, then $\int_0^T F(s) ds$ would collapse to a pure boundary term and the pathwise claim would become trivial. Therefore, the claim $L(T) = 0$ remains empirically falsifiable rather than analytically guaranteed.

The telescoping hazard: empirical observables must be non-conservative and path-dependent; otherwise the ledger collapses to a boundary term $S(T) - S(0)$.

Methodological Note on Integration: To prevent algebraic collapse, the Law mandates that the empirical observables z_k cannot be conservative state variables. They must represent non-conservative, path-dependent physiological regulatory-burden or regulatory-activity proxies. Allostatic-load style measures are one candidate family of proxies, but because allostatic load remains operationally heterogeneous across the literature, no single biomarker algorithm is treated as uniquely canonical here. Because empirical measurements arise from discrete physical telemetry, the empirical estimator is constructed as a discrete Riemann sum on these path-dependent contribution values. To mitigate discretization errors and ensure non-conservativeness, $z_k(t_i)$ must represent path-dependent regulatory activity that cannot be represented, at the measurement scale used, as the exact derivative of a conservative state function (e.g., autonomic load markers or endocrine stress dynamics serving as empirical indicators of metabolic regulatory load). Preregistered pipelines incorporate stochastic noise models (e.g., telemetry variance) and Monte Carlo simulations with preregistered replication counts chosen for adequate Monte Carlo precision to characterize estimator behavior under the observation model and to assess whether the chosen observables remain non-conservative and path-sensitive under realistic noise and sampling conditions. Preregistered tests will use path-dependence diagnostics together with dependence-preserving surrogate or block-resampling procedures, rather than naive trajectory permutation alone, to assess whether the resulting contribution sum is empirically compatible, at the measurement scale used, with reduction to a conservative state difference $S(T) - S(0)$. The decision rule relies on dependence-aware reconstruction error and predictive failure of conservative state-function models rather than a two-tailed t-test for mean differences.

$$HCI(t_i) = \sum_k w_k z_k(t_i)$$

Here $w_k \geq 0$ are preregistered weights and $z_k(t_i)$ are preregistered signed, non-conservative physiological contribution channels derived from intensive telemetry relative to preregistered anchors or subject-specific baselines defined by preregistered rules; the sign convention resides in $z_k(t_i)$, not in post hoc changes to w_k . The empirical estimator of the phenomenological ledger:

$$L(t) = \int_0^t g_\theta(s) F(s) ds$$

is constructed from physiological telemetry as:

$$\dot{L}(t) = \sum HCI(t_i) \Delta t_i$$

Because the constituent observables $z_k(t_i)$ are preregistered non-conservative, path-dependent quantities, this discrete sum cannot be algebraically reduced to a boundary difference $S(T) - S(0)$ through telescoping.

Structural safeguards:

- **Preregistration:** Weights w_k , anchors or baseline rules, and preprocessing pipelines are fixed prior to data collection.
- **Invariance:** For cross-group comparisons, LoF requires empirical support for at least metric invariance of the preregistered channel-level observation model underlying HCI across relevant demographics; scalar invariance is required for cross-group level comparisons. When subject-specific baselines are used in channel construction, preregistered harmonization or hierarchical measurement modeling must preserve the invariance target for the intended comparison. Preregistered tests (e.g., multi-group confirmatory factor analysis or closely related latent-variable checks on the channel-level observation structure underlying HCI) must confirm the relevant level of invariance in diverse global cohorts (e.g., non-WEIRD samples), incorporating cultural covariates (e.g., individualism scores per Hofstede et al., 2010). Failures restrict claims to the specific comparison class supported by the data. Any recalibration should be treated as exploratory and confirmed only in a new preregistered dataset.
- **No post hoc tuning:** adjustments to w_k after observing outcomes are prohibited.

Neutrality must emerge from system dynamics rather than algebraic identity.

Defining “Death of Mind” via retrospective irreversibility requires future information, which violates real-time causality. Instead, the endpoint T is defined as a stopping time with respect to the observation filtration $\mathcal{Y}_t$:

$$T = \inf_{t \geq 0} \{ t: \theta(t) \leq \theta_0 \}$$

Here, $\theta(t)$ is a preregistered proxy for the unity of conscious access, anchored to perturbational-complexity and related cortical-integration markers (Casali et al., 2013); where feasible, fast TMS-EEG perturbational complexity estimators (Comolatti et al., 2019) may contribute, but any extension beyond disorders-of-consciousness or anesthesia-linked settings to naturalistic terminal trajectories should be treated as provisional and separately validated. Operationally, the stopping event itself is determined only by the preregistered unity-proxy channels rather than by the ledger estimator. The event $\{T \leq t\}$ is $\mathcal{Y}_t$ -measurable.

The observable proxy $\theta(t)$ is hypothesized to decline, on average, as the latent biological reserve $U(t)$ depletes. Short-term reversals are permitted, but preregistered longitudinal models must demonstrate a consistent negative association between $\theta(t)$ and reserve depletion, or equivalently a positive association between $\theta(t)$ and validated reserve proxies, on the time scale relevant to the stopping rule. This should be assessed using mixed-effects or semi-parametric models (e.g., GAMs) rather than by a single hard monotonicity threshold. If the negative association is weak or unstable, endpoint claims should be restricted to validated subpopulations, and collapse of $\theta(t)$ should be treated only as a provisional proxy indicator of reserve depletion. Death of Mind

corresponds to collapse of unified experience capacity as tracked by $\theta(t)$ and operationalized by the threshold crossing $\theta(t) \leq \theta_0$, not by the content of the experience captured by L .

Ledger gating is explicit. Define the unity gate: $g\theta(t) = 1\{\theta(t) > \theta_0\}$. In the phenomenological interpretation, the ledger accrues only while the gate is open:

$$L(t) = \int_0^t g_\theta(s) F(s) ds$$

This blocks post-collapse phenomenological integration and should be read as the gated phenomenological representation of the ledger rather than as a second primitive stochastic definition.

The Logical Fork Lemma: If T is exogenous to the ledger dynamics and, conditional on stopping-time information and any preregistered covariates that do not algebraically encode the ledger, the law of $L(T)$ remains nondegenerate and absolutely continuous with respect to Lebesgue measure, then $P(L(T) = 0) = 0$. For affective dynamics featuring discrete events (e.g., trauma), the same obstruction extends to jump-diffusion models only when the conditional law of $L(T)$ remains nondegenerate and absolutely continuous with respect to Lebesgue measure; preregistered simulations using pilot-calibrated jump intensities and amplitudes should test whether that condition is plausibly preserved under discontinuous perturbations (Merton, 1976). To recover the latent strong-form condition $P(L(T) = 0) = 1$, at least one lemma condition must fail (e.g., via terminal degeneracy induced by reserve collapse in the multiplicative model $L(t) = U(t)Y(t)$). Closure cannot arise from randomness alone under the exogeneity and continuous-density assumptions above unless state-coupling or degeneracy is present.

The core reserve-coupled construction satisfies this via terminal degeneracy ($U(t) \rightarrow 0$), committing the model to state-coupled termination. This coupling is broadly consistent with age-related neuromodulatory models in which declining plasticity and dopaminergic flexibility are associated with reduced affective variability (Li & Sikström, 2002; Bäckman et al., 2010), and is empirically testable via preregistered lead-lag models on HCI and $\theta(t)$, including Vector Autoregression (VAR) where approximate stationarity and regular sampling are defensible, and state-space or continuous-time analogues when sampling is irregular or strong nonstationarity is expected. Lead-lag diagnostics such as impulse-response analysis or their state-space analogues must test whether deterioration in $\theta(t)$, or in independently validated reserve proxies linked to reserve depletion, statistically precedes or tightly tracks contraction of rolling ledger variance under preregistered lead-lag criteria. This lead-lag structure is an empirical signature rather than a stand-alone proof of mechanism and should be tested against shared-cause alternatives such as age, disease burden, medication exposure, and general frailty. Lag structure should be preregistered in units consistent with the sampling cadence and justified by pilot data or domain-specific temporal considerations, including plausible seasonal, longitudinal, or reserve-linked temporal effects when relevant.

Edge Cases and Abrupt Termination: Abrupt, exogenously forced terminations (e.g., sudden accidents) belong to a distinct competing-risk regime and are not part of the first-pass falsification package for the naturalistic reserve-decay architecture. Large nonlethal late-life perturbations that introduce substantial jump-like imbalance before reserve has entered the low-reserve terminal regime should likewise be treated as a separately modeled perturbation regime rather than as a clean first-pass test of the diffusion-dominated terminal-layer approximation. If reserve-coupled anticipatory dynamics exist in such cases, they should appear only as preregistered deviations in reserve or stress proxies before a subset of abrupt terminations or late perturbation events. Failure to detect such signatures in these auxiliary regimes weakens only those extensions, not the core naturalistic-decay architecture. For comas or other prolonged reductions of unified conscious access, LoF predicts that the unity gate suppresses or prevents further ledger accrual once $\theta(t)$ falls below threshold. For flickering $\theta(t)$, including transient lucidity-like rebounds near the endpoint, a preregistered smoothed gate or dwell-time rule may be used to reduce artifactual switching and to prevent brief rebounds from redefining the stopping event. Any developmental or pediatric extension requires separate validation and should not be assumed from the adult stopping-rule framework.

A frequent objection from computational disciplines is that reinforcement learning or homeostatic regulation already explains affective dynamics. The key distinction for the present comparison is not a generic claim about all reinforcement-learning variants, but the specific requirement of pathwise closure and horizon-sensitive admissibility under a realized finite-horizon trajectory. Standard reinforcement learning optimizes expected cumulative reward over an ensemble of trajectories, e.g., $\operatorname{argmax}_{\pi} E_{\pi}[\int_0^T r(t) dt]$. The Law of Fairness avoids tautological closure by ensuring that the stopping rule (collapse of the Unity Index) and the ledger estimator (HCI) are not defined from the same physiological measurements. Residual cross-channel dependence may exist biologically, but it must be modeled explicitly rather than collapsed into a single signal. This prevents the boundary condition from being defined by the same signal used to measure the ledger. This allows terminal variance compression to emerge as a testable physical constraint signature, rather than a tautological restatement of the stopping rule.

While standard ensemble-optimizing RL permits extreme pathwise outcomes, LoF instead posits strict realized-trajectory closure under a non-ergodic process-level architecture, while still allowing subprocess ergodicity within neural populations. This contrasts with standard RL and has partial formal overlap with safe-RL variants (Garcia & Fernández, 2015), but the LoF claim is stronger because it imposes realized pathwise closure rather than expected-return optimization with safety penalties or constraints alone. Preregistered simulations should test whether unconstrained RL permits $|L(T)| > K$ in a non-negligible fraction of simulated trajectories, whereas in the LoF construction suppression arises from state-coupled multiplicative contraction of the diffusion coefficient $\sigma U(t)$ and from horizon-dependent admissibility constraints rather than from expectation optimization alone. Extending cognitive effort models (Shenhav et al.,

2013), this is testable via priced-constraint experiments using simulation-based power analysis and bootstrap-based uncertainty checks for robustness.

Along a realized path in a non-ergodic system, the time average need not converge to the ensemble expectation. Because the latent strong form posits pathwise closure $L(T) = 0$, the governing dynamics cannot be modeled as an unconstrained expectation optimizer alone. As the horizon shrinks, dispersion must be actively suppressed and feasible actions must tilt toward compensable trajectories.

Empirical Test, Priced Constraint: Experimental tasks introduce choices where high-compensability actions carry a small immediate cost. Prediction: as the conditional horizon $H(t)$ shrinks (equivalently, as $H(t)^{-1}$ increases), willingness to incur an immediate cost for closure increases. Models must control for hyperbolic discounting, baseline risk, and loss aversion via preregistered covariates (e.g., Prospect Theory parameters per Kahneman & Tversky, 1979) to isolate Φ effects. The coefficient on $\Phi \times H(t)^{-1}$ must be significantly positive under preregistered baselines, with multicollinearity checked for all terms, including interactions. If needed, regularized estimators may be used with tuning fixed by cross-validation or preregistration. Sample size and minimum detectable effects should be determined by preregistered simulation-based power analysis matched to the actual task design, dependence structure, and correction for multiple testing. Extending Shenhav et al. (2013), preregistered fMRI tasks should test whether Φ -related variance remains in control hubs after controlling for standard predictors; repeated null results under adequate power would falsify this mechanism-level claim.

We construct a stochastic process consistent with the latent strong-form idealization without mathematical tautology. The biological reserve $U(t)$ is defined as residual biological plasticity with a hypothesized physical substrate and preregistered empirical reserve proxies. Over naturalistic trajectories, cumulative metabolic burden, allostatic-load-related wear, and age-linked reductions in plasticity are hypothesized to reduce reserve. In the CNS, this may manifest as reduced synaptic plasticity or broader reductions in neural flexibility. $U(t)$ is modeled as a latent reserve factor whose candidate empirical realizations include autonomic flexibility, endocrine regulation, neural complexity, and composite reserve indices shaped by cumulative wear. The reserve interpretation is empirical rather than definitional and should be rejected if preregistered reserve proxies fail to improve prediction beyond simpler baselines such as age, frailty, or generic decay models. Where feasible, preregistered reserve proxies should also avoid algebraic identity with either HCI or $\theta(t)$, with any unavoidable overlap modeled explicitly rather than treated as independence by default.

On naturalistic preterminal trajectories, $U(t)$ is treated as adapted, of finite variation, and strictly positive for $t < T$; we model net plasticity decline as a depletion process:

$$dU(t) = -v(t) dt$$

where $v(t) > 0$ is an adapted process representing the net rate of biological reserve depletion. To ensure that reserve depletion enters the terminal regime in finite time, we impose the maintained terminal-layer assumption that $v(t) \geq \varepsilon > 0$ on the relevant preterminal interval, so that $U(t)$ reaches sufficiently low levels in finite time when $U(0)$ is finite. Finite-time occurrence of T then depends on the separately preregistered stopping-rule relationship between $\theta(t)$ and reserve depletion, rather than following from $U(t)$ alone. If validated reserve-factor estimates fail to support a strictly positive terminal lower-envelope for $v(t)$, the drift-dominance claim should be treated as unsupported even if weaker concentration tests remain evaluable.

This finite-variation reserve law is the reference architecture for the present concentration result; stochastic reserve variants with correlated noise or partial recovery are admissible future extensions but are not assumed in the current formal result. The latent strong form does not require any single empirical reserve proxy to literally attain zero at the observed endpoint; empirically, the relevant question is whether preregistered reserve proxies enter the predicted low-reserve terminal regime and covary with the concentration signatures of the model. We acknowledge that some reserve proxies or aging-related markers may be partially modifiable under intervention, although the extent and mechanism depend on the specific marker. Trajectories involving interventions that materially alter reserve proxies are treated as a separate regime requiring explicit modeling; the strong-form boundary condition is defined for naturalistic trajectories without such regime shifts.

$$T = \inf_{t \geq 0} \{ t: \theta(t) \leq \theta_0 \}$$

where $\theta(t)$ is a preregistered proxy for unity of conscious access, derived from physiological channels that are not algebraically or instrumentally identical to those defining HCI (e.g., EEG perturbational complexity indices vs. autonomic HRV/cortisol markers). Preregistration must include empirical dependence diagnostics (e.g., Canonical Correlation Analysis and mutual-information estimation calibrated by simulation; see Kraskov et al., 2004) to quantify residual preterminal dependence among $\theta(t)$, HCI, and the preregistered reserve-proxy package, and to assess whether endpoint definition and reserve operationalization remain operationally distinct from ledger measurement. Where these constructs draw from overlapping physiological batteries, the shared structure must be modeled explicitly within the state-space observation model rather than treated as negligible. This definitional mapping is standard in biophysical modeling and preserves the dynamical role of the stochastic process. The reserve-coupled stochastic dynamics still govern the ledger trajectory and the variance-compression signature, while the endpoint T remains separately defined through $\theta(t)$.

The multiplicative structure does not establish that biological systems obey the Law of Fairness. Its role is strictly mathematical: it demonstrates that exact pathwise closure is dynamically achievable in a state-coupled stochastic system without defining the stopping rule by the ledger itself and without invoking backward causation. Empirically, this structural interpretation is weakened if preregistered analyses fail to detect the predicted association between collapse of $\theta(t)$

and terminal variance compression. Falsification does not test the algebra; it tests whether actual biological systems exhibit the predicted physical signatures of this constraint (e.g., drift dominance, variance compression).

Let $Y(t)$ be defined by $dY(t) = \sigma dW(t)$ with an arbitrary initial state $Y(0) = Y_0$. We define the ledger by multiplicative coupling:

$$L(t) = U(t)Y(t)$$

The stochastic construction $L(t) = U(t)Y(t)$ should therefore be interpreted as one formal realization of the same underlying ledger concept at the level of modeled consequences rather than as a pointwise identification of $F(t)$ with a classical derivative of $U(t)Y(t)$. In the phenomenological representation the ledger is defined by the unity-gated time integral:

$$L(t) = \int_0^t g_\theta(s) F(s) ds$$

while the reserve-coupled stochastic dynamics provide one mathematically explicit architecture through which the terminal condition $L(T) = 0$ can arise. This coupling is broadly consistent with hypotheses in which higher-plasticity states can support larger affective fluctuations, whereas aging-associated reductions in neuromodulatory flexibility can compress them (Li & Sikström, 2002; Bäckman et al., 2010).

Within the strong-form structural idealization, exact closure becomes dynamically available under the imposed biological decay constraint together with multiplicative coupling. Because T is a finite stochastic stopping time and $Y(t)$ has continuous sample paths under the reference diffusion, $Y(T)$ is finite almost surely. If heavier-tailed càdlàg alternatives are explored, pathwise finiteness at finite stopping times should be verified separately under the chosen process class, even when higher moments may fail to exist. If the stopping manifold aligns with terminal reserve collapse so that $U(T) = 0$ in the latent process, then $L(T) = U(T)Y(T) = 0$ follows almost surely. Empirically, however, this exact equality is not observed directly; the measurable targets are terminal concentration near zero and equivalence-band neutrality in the estimator $\hat{L}(T)$.

On any compact preterminal interval $[0, T - \delta]$ with $\delta > 0$, Itô's lemma yields dynamics; the terminal statements below are limiting claims as $t \rightarrow T^-$ rather than assertions of regularity at the singular endpoint:

$$dL(t) = -\left(\frac{v(t)}{U(t)}\right) L(t) dt + \sigma U(t) dW(t)$$

Here $v(t)$ is adapted but treated as non-diffusive for simplicity. Because $U(t)$ is modeled as a finite-variation process, no quadratic cross-variation term appears in the Itô expansion for $L(t)$.

For a fixed reserve path $U(\cdot)$ of finite variation (treated as exogenous in the conditional Fokker–Planck formulation), the conditional density $p(l, t | U(\cdot))$ satisfies the Fokker–Planck equation:

$$\partial_t p = \left(\frac{v(t)}{U(t)} l p \right)_l + \frac{1}{2} (\sigma^2 U(t)^2 p)_{ll}$$

Because the diffusion coefficient $\sigma U(t)$ vanishes as $U(t) \rightarrow 0$, the reserve-coupled ledger concentrates near $L = 0$ as the stopping time approaches, without imposing an absorbing boundary at $L = 0$ prior to T . More precisely, under the core reserve-coupled construction $L(t) = U(t)Y(t)$, if $U(t)$ is adapted, nonnegative, and of finite variation, $U(t) \rightarrow 0$ as $t \rightarrow T^-$ almost surely, and $\sup_{\{t < T\}} E|Y(t)|^2 < \infty$, then for every fixed $\delta > 0$ one has $P(|L(t)| > \delta) \rightarrow 0$ as $t \rightarrow T^-$. Equivalently, $L(t)$ converges to 0 in probability in the terminal layer.

Conditioned on a fixed reserve path $U(\cdot)$ in the exogenous-path formulation $E[L(t)^2 | U(\cdot)] = U(t)^2 E[Y(t)^2 | U(\cdot)]$; in the independent-reference-diffusion construction this reduces to $U(t)^2 E[Y(t)^2]$. Hence the conditional second moment also tends to 0 whenever $\sup_{\{t < T\}} E[|Y(t)|^2 | U(\cdot)] < \infty$; in the independent-reference-diffusion construction this is implied by $\sup_{\{t < T\}} E|Y(t)|^2 < \infty$.

This concentration result does not by itself prove that biological systems instantiate the reserve-coupled structure; rather, it shows that if the core latent architecture is correct, terminal concentration near neutrality is a mathematically necessary conditional signature of the model, with unconditional cross-sectional contraction requiring appropriate conditioning or stratification on reserve, horizon, or a comparable terminal regime. As a numerical sensitivity analysis only, Robin boundaries with preregistered absorption rates may be used to model gradual collapse, while reflecting boundaries at large $|L|$ approximate an effectively unbounded domain in numerical schemes. Mass concentration near $L = 0$ is evaluated through preregistered reserve-stratified or terminal-regime-conditioned concentration diagnostics such as rolling-variance contraction, shrinking quantile spread, and model comparison against unconstrained baselines; equivalence testing is reserved for the separate empirical target of terminal neutrality in $\tilde{L}(T)$. Numerical solutions (e.g., finite-difference methods, with numerical stability checked via the Courant–Friedrichs–Lewy stability condition for the chosen finite difference discretization) illustrate the expected concentration of the conditional density near zero as $U(t) \rightarrow 0$ and should be treated as consistency checks rather than empirical confirmation by themselves.

As the biological reserve $U(t) \rightarrow 0$, two empirical signatures emerge:

- **Drift Dominance:** Under the maintained assumption that $v(t)$ remains bounded below by a positive constant in the terminal layer, the mean-reversion rate $v(t)/U(t)$ diverges as $U(t) \rightarrow 0$, generating increasingly strong mean-reversion pressure.

- **Variance Compression:** The diffusion coefficient $\sigma U(t)$ vanishes, suppressing stochastic excursions and driving the probability distribution toward strong concentration near zero as the terminal boundary is approached.

Predictions must account for sensor noise near the endpoint via preregistered state-estimation pipelines such as Kalman, extended or unscented Kalman, particle-filter, or closely related methods, with noise parameters estimated from calibration data or jointly inferred under the observation model. Because terminal telemetry may also be missing in a nonrandom way due to device dropout, sedation, agitation, or care transitions, preregistered missing-data analyses should include explicit nonrandom-missingness sensitivity models, such as shared-parameter, selection-model, or pattern-mixture perturbations consistent with the observation model. If the variance-compression result disappears under plausible missingness stress tests, empirical support is weakened accordingly.

The system cannot know the realized T in advance. Instead, on the preterminal interval it operates on a filtered conditional horizon $H(t) = E[(T - t)^+ | \mathcal{Y}_t]$, or its preregistered state-space approximation, so the resulting problem is most naturally viewed as an exit-time stochastic optimal-control problem rather than regulation over a fixed deterministic horizon. In a constrained optimal-control formulation, $\lambda(t)$ represents the canonical marginal value-gradient penalty on residual imbalance.

The shadow price $\lambda(t)$ increases as the filtered horizon $H(t)$ shrinks and may exhibit approximate inverse-horizon scaling in simplified linear-quadratic control approximations; the exact scaling depends on the assumed cost functional. For larger σ regimes, and only after a separate preregistered cost functional has been specified, the constrained optimal-control problem may be written in Hamilton-Jacobi-Bellman (HJB) form and represented in forward-backward stochastic differential equation language, where the forward state evolves the reserve-coupled ledger and the backward costate propagates the consequences of the terminal boundary condition into present-time control pressure. Because the terminal neutrality condition acts as a hard boundary in representative reduced-form control formulations, the value function can develop a steep terminal layer as the endpoint approaches; the strength and exact form of that layer depend on the separately specified objective functional and regularity assumptions. In a fully observed reduction, $\lambda(t)$ may be identified, up to sign convention, with $\partial V / \partial L$; under partial observation the corresponding object is the gradient of the value functional on the filtered sufficient state or belief state. In either formulation, the control pressure associated with residual imbalance increases sharply near closure.

Assuming $E[L(t) | \mathcal{Y}_t] \approx 0$, the expected required compensation scales approximately as:

$$F_{req} \approx -E[L(t) | \mathcal{Y}_t] / H(t)$$

under a Brownian small-variance assumption, with non-Markovian extensions via filtered-state or recurrent estimators for $H(t)$. The admissible set $A_t \subset A$ is defined as the set of actions preserving closure feasibility under bounded compensability. In the most direct control-theoretic form, feasibility requires $|L(t)| \leq H(t) F_{max}(U(t), m(t))$, where $m(t)$ denotes the currently available channel state and $F_{max}(U(t), m(t))$ is the preregistered bound on attainable compensatory authority given reserve and currently available channels.

The reduced-form envelope $|L(t)| \leq c U(t)^\alpha$ is then treated as an empirical summary of that deeper feasibility structure rather than as an unexplained axiom (cf. Aubin, 2009). Under partial observation, admissibility is operationally assessed on filtered state estimates or belief states; a natural operational form is the chance constraint $\pi(t)[|L(t)| \leq c U(t)^\alpha] \geq 1 - \varepsilon$, and actions that render closure infeasible are removed or strongly suppressed.

Compensability $\Phi(a | \pi(t))$ quantifies the expected change in closure feasibility under partial observation, formally defined via preregistered belief-MDP simulations as the expected change in filtered closure feasibility given action a and current belief state $\pi(t)$. The shadow price $\lambda(t)$, formally defined as the marginal value-gradient penalty associated with preserving terminal admissibility, rises primarily through shrinking $H(t)$, tightening feasibility bounds, and increasing steering cost. The Viability Ratio $\Psi(t) = |L(t)| / U(t)$ should therefore be interpreted as a reserve-normalized latent-amplitude factor rather than as a generically diverging urgency variable. A risk-sensitive interpretation may be layered onto this structure as a secondary refinement (cf. Biswas & Borkar, 2023), but the primary drivers of pruning are filtered feasibility, shrinking horizon, and bounded compensatory authority.

Empirical Horizon Scaling Predictions: Operationally, admissible menu counts are predicted to decrease approximately with inverse horizon $H(t)^{-1}$ as the filtered horizon $H(t)$ shrinks, and to exhibit overdispersion, tested via preregistered Negative Binomial generalized linear mixed models with model fit compared to Poisson alternatives. Neurobiologically, the prediction is that after controlling for utility, conflict, arousal, and fatigue, control hubs such as rIFG and dACC, building on established models of time-pressure threshold adjustments (Shenhav et al., 2013), retain residual explanatory signal attributable to Φ and $\Phi \times H(t)^{-1}$ interaction terms. Absence of this residual signal under adequate preregistered statistical power, with sample size determined by design-specific simulation or power analysis, constitutes a mechanistic strike against the QS hypothesis.

In waking life, external constraints may prevent behavioral compensation (e.g., imprisonment, systemic paralysis). If LoF holds as an almost-sure path constraint in the latent process, one possible biological pathway could involve internal neural dynamics capable of altering affective balance offline. One auxiliary biological mechanism consistent with LoF is that a subset of sleep-dependent neural dynamics may contribute to compensatory affective rebalancing under some conditions. In that auxiliary regime, following negative waking ledger accumulation L_{wake} , REM

affect is predicted to shift toward neutrality or mastery-themed states across successive REM periods. This is treated as a discriminable mechanism-level hypothesis rather than as a load-bearing foundation of the core model.

Mechanism Framing: NREM slow-wave activity has been hypothesized to support synaptic and metabolic restoration. By contrast, one auxiliary offline affective-balancing hypothesis in the present model focuses primarily on REM sleep, while alternative sleep-dependent emotional-processing accounts remain live competitors. Healthy REM suppresses noradrenergic (norepinephrine) tone, permitting affective processing of highly valent memories without triggering the full autonomic stress response. This may contribute to affective rebalancing during sleep, potentially manifesting as shifts toward more neutral or mastery-themed dream affect. This hypothesis extends emotion-regulation theories (Cartwright et al., 1998) and is broadly consistent with REM neurobiology and protoconsciousness-oriented accounts of REM function (Hobson, 2009).

Empirical Falsifiers: Morning dream reports are coded for valence alongside polysomnography. The mechanism is falsified if preregistered mixed-effects models fail to detect a within-night progression in REM affect following negative waking ledger accumulation L_{wake} , such that earlier REM periods are more negative and later REM periods shift toward neutrality or mastery-themed content. Analyses must control for recall and non-affective REM functions (e.g., memory consolidation). Power and coding-reliability targets should be determined by preregistered design-specific analysis plans matched to the sampling cadence, scoring protocol, dependence structure, and measurement model of the study. The predicted signature is a within-night shift toward less negative and more mastery-themed REM affect under healthy sleep architecture.

Biological Caveat: Pathological states such as PTSD or chronic depression involve documented REM anomalies and heightened amygdala reactivity linked to noradrenergic dysregulation (Germain et al., 2008). Rather than dismissing these anomalies as mere mechanistic exceptions, the framework treats them as hard boundary tests. If preregistered mixed-effects models fail to detect the predicted within-night rebalancing pattern in REM affect under adequate statistical power, the mechanism is rejected. While hypothesized as a modifiable vulnerability factor, bidirectional associations between PTSD and sleep disturbances are acknowledged; preregistered longitudinal designs should assess temporal asymmetry and lagged directional structure using cross-lagged or closely related longitudinal models.

A biological motivation for the reserve term, rather than a derivation of the boundary condition itself, is that sustained extreme affective states may be metabolically costly and physiologically destabilizing. Persistent high-magnitude affective states are hypothesized to correlate with sustained regulatory mismatch and, in many pathological regimes, prolonged autonomic activation. Maintaining such states may require sustained regulatory effort, plausibly contributing to increased metabolic load and depletion of biological plasticity $U(t)$.

The viability envelope and related feasibility conditions should not be interpreted as quantities explicitly computed by the brain. They are analytical summaries of a physical constraint imposed by declining biological reserve. As reserve $U(t)$ collapses, the metabolic and regulatory cost of sustaining large affective deviations rises sharply. Control hubs such as dACC and rIFG need not literally compute a symbolic future horizon; instead, horizon-sensitive behavior can emerge from filtered state estimation and local control updates, while $H(t)$ functions as a researcher-level sufficient statistic summarizing how local control policies become more restrictive as reserve and channel feasibility decline. The mathematical boundary $|L(t)| \leq H(t) F_{\max}(U(t), m(t))$ is thus a compact analytical representation of this constraint rather than a literal symbolic computation performed by the organism.

No teleological or acausal mechanism is required. Sustained high-magnitude affective imbalance and high-variance affective trajectories are hypothesized to destabilize the organism metabolically. In a stochastic-control interpretation, maintaining trajectories away from neutrality can be represented as requiring increasing control burden. Under standard controlled-diffusion assumptions and the usual integrability conditions for measure change, this burden can be represented by a change of path measure from an unconstrained diffusion law P to a controlled law Q . Girsanov’s theorem provides the measure-change framework; in entropy-regularized control formulations, the associated steering burden can then be quantified by a relative-entropy cost such as $KL(Q || P)$. This gives the control burden associated with $\lambda(t)$ a path-space interpretation as the marginal cost of deforming the uncontrolled law toward terminally neutral admissible trajectories. This control-cost interpretation is mathematically precise within stochastic control theory and should be distinguished from broader thermodynamic analogies, which should remain outside the core model unless the relevant stochastic current and entropy-production observables are explicitly defined.

Drawing on established indices of allostatic load (Juster et al., 2010) and models of allostatic-load energetics (Bobba-Alves et al., 2022), high-variance trajectories are therefore hypothesized to be constrained by declining systemic metabolic capacity. Evidence from allostatic-load and stress-biology research suggests that sustained regulatory burden may be associated with increased metabolic load and with broader aging-related biological burden (Juster et al., 2010; Bobba-Alves et al., 2022). In the present framework, this possibility is most cleanly tested via preregistered genetically informed or longitudinal designs linking burden indices derived from preregistered affective-variance and reserve proxies to biological aging markers (e.g., epigenetic clock measures; Horvath, 2013) under a prespecified power analysis and measurement model.

This proposed metabolic penalty is consistent with allostatic energetics accounts and may also be interpreted within broader biodemographic aging frameworks in which declining reserve increases vulnerability to terminal failure (Vaupel, 2010). Within the present framework, systems that chronically fail to regulate the time-integrated affective load may, on average, become more vulnerable to earlier terminal failure under metabolic constraints. Constraint-binding control architectures are compatible with blind evolutionary selection pressures acting on physiological

stability. Limiting sustained high-magnitude affective imbalance is biologically consistent with preserving organized physiological regulation under metabolic constraints. These considerations motivate the reserve-depletion and bounded-compensation terms in the model; they do not by themselves establish the neutrality boundary, which remains a distinct hypothesis to be tested against simpler homeostatic, frailty, and aging baselines. Standard homeostatic models, such as Ornstein-Uhlenbeck dynamics, define affect as:

$$dF(t) = -\kappa F(t) dt + \sigma dW(t)$$

Although this process locally regresses to the mean, the integrated ledger $L(t) = \int_0^t F(s) ds$ generally accumulates variance over long horizons when no terminal boundary constraint is imposed; for zero initial condition $F(0) = 0$, the exact closed-form:

$$\text{Var}(L(t)) = \frac{\sigma^2}{2\kappa^3} (2\kappa t - 3 + 4e^{-\kappa t} - e^{-2\kappa t})$$

grows asymptotically linearly with slope σ^2/κ^2 for large t . For regularly sampled HCI series, κ may be approximated from autocorrelation via AR(1) fits; under irregular sampling or strong nonstationarity, κ should instead be estimated in a continuous-time or state-space formulation. For sufficiently large κ relative to the observational horizon T , the variance growth slows but does not vanish, so comparative evaluation should rely on preregistered model comparison rather than on the existence of mean reversion alone. As a result, passive random surpluses or deficits at T remain possible with non-zero probability under standard modeling.

Because the latent strong form posits $L(T) = 0$ as a terminal boundary condition, the process cannot remain an unconstrained diffusion. It must behave like a state-coupled constrained or controlled diffusion capable of producing terminal concentration and, under the strong-form idealization, pathwise closure. The critical empirical test emerges as the conditional horizon $H(t)$ approaches zero (e.g., in hospice contexts). A central prediction of this framework is that reserve-stratified cross-sectional ledger variance, and within-trajectory rolling ledger variance, contract as the process approaches the terminal stopping time T ; in the exogenous-path formulation of the coupled process $L(t) = U(t)Y(t)$, the directly derived statement is the conditional result:

$$\lim_{t \rightarrow T^-} \text{Var}(L(t) | U(\cdot)) = 0$$

While theoretically derived from the reserve-coupled diffusion construction, this conditional boundary behavior is only qualitatively aligned with gerontological observations such as loss of physiological complexity and reduced heart-rate variability in aging and frailty (Chaves et al., 2008; Lipsitz & Goldberger, 1992), not directly confirmed by them. Ledger variance is proxied via HCI rolling variances, with contraction tested primarily through preregistered variance-function or mixed-effects models, while supplementary robust variance diagnostics such as Levene's test may be reported on preregistered reserve-proxy-stratified bins or bins defined on estimated $U(t)$, explicitly controlling for confounds such as polypharmacy through preregistered matching or

covariate-adjustment strategies using age, comorbidities, medication burden, and ATC medication codes, or by adjustment for medication classes (e.g., opioids, beta-blockers).

In the period preceding natural death, organisms can exhibit measurable reductions in cognitive and autonomic dynamism, with Heart Rate Variability (HRV) in some settings showing reduced variability or complexity that may serve as an imperfect reserve-related signal rather than as a direct proxy for affective flexibility. The mathematical variance compression of the ledger formalizes the predicted contraction of the latent ledger process, while autonomic and endocrine measures provide noisy empirical indicators of rising regulatory cost under declining reserve. We employ Two One-Sided Tests (TOST) to evaluate terminal neutrality.

Let $\hat{L}(T)$ denote the empirical estimate of the lifetime ledger $L(T)$.

- **Neutrality Supported:** the 90% confidence interval (corresponding to $\alpha = 0.05$ in the Two One-Sided Test procedure) of the empirical ledger estimate $\hat{L}(T)$ lies entirely within the preregistered bounds $[-K, +K]$.
- **Sample-level imbalance signal:** If the 90% confidence interval for the estimator $\hat{L}(T)$ lies wholly above $+K$ or wholly below $-K$, that sample provides directional evidence of terminal imbalance. Theory-level falsification still requires repeated preregistered violations across well-measured cohorts together with the cohort-level exceedance logic specified below.
- **Inconclusive:** If the 90% confidence interval overlaps one or both equivalence bounds without full containment, that sample does not support neutrality but also does not by itself establish terminal imbalance.

To ensure additional statistical rigor, TOST is supplemented with Bayes factors computed under a preregistered weakly informative prior family together with preregistered sensitivity analysis over reasonable prior scales. Under simple homoskedastic observation-noise benchmarks with fixed sampling assumptions, the propagated uncertainty of the integrated estimator can scale on the order of $\sqrt{T}$; more generally, K should be derived from the full observation model together with a preregistered smallest practically interpretable terminal departure from neutrality at the chosen measurement scale, rather than fixed by pilot convenience or an assumed universal scaling law.

To account for potential skewness and serial dependence in path-dependent non-Gaussian physiological sums, confidence intervals should be estimated using preregistered dependence-aware resampling or posterior-simulation procedures consistent with the observation model, with replication counts chosen for adequate Monte Carlo precision. For highly non-Gaussian $\hat{L}(T)$, Bayesian evidence assessment should be reported under that same preregistered prior family and its sensitivity analysis rather than under a single untuned default scale. Variance comparisons should be interpreted primarily through the preregistered variance-function or mixed-effects estimates, with Levene’s test reported only as a supplementary diagnostic rather than as a primary inferential test.

A $\text{BF}_{01} > 30$ is required as strong evidence favoring the neutrality model relative to the preregistered imbalance alternative. Conversely, a $\text{BF}_{10} > 30$ in favor of terminal imbalance constitutes a preregistered dataset-level falsifier and contributes to theory-level falsification under the cohort logic below.

The following outcomes constitute preregistered core falsifiers or hard restrictions on inference, as applicable. Where families of frequentist tests are used, multiplicity control must be preregistered, and theory-level claims should rely on a preregistered cross-domain evidence-aggregation rule rather than on any single supportive domain in isolation:

- **Terminal imbalance:** repeated well-measured streams end with confidence intervals strictly outside the neutrality band, with the preregistered cohort-level exceedance threshold determined from the expected error rate under neutrality and the measurement model rather than fixed post hoc.
- **Variance expansion:** preregistered longitudinal variance-function or mixed-effects analyses show expanding ledger variance near T , with supporting diagnostics across preregistered time bins, violating the compression signature.
- **Invariance breakdown:** if the preregistered channel-level observation model underlying HCI fails the level of invariance required for a given cross-group comparison near death, that comparison is not interpretable at the intended level.

Operational Summary: The empirical ledger estimator $\hat{L}(T)$ is constructed by integrating a preregistered Hedonic Composite Index (HCI) derived from longitudinal multimodal physiological telemetry and related preregistered contribution channels. The endpoint T is defined as a stopping time on a preregistered unity threshold $\theta(t)$, not by solving $L(T) = 0$. Neutrality is tested using a preregistered $\pm K$ equivalence band with TOST and Bayes factor confirmation.

Scientific integrity is preserved by freezing the endpoint definition T , the weights w_k , the anchors or baseline rules, the preprocessing pipeline, and the error budget K before data collection. If the observation model fails preregistered reliability criteria, fails discriminant-validity checks against preregistered nuisance constructs, fails the level of invariance required for the intended comparison, or fails non-conservative path-dependence requirements, the affected analysis provides no support for LoF at that level and should not be pooled as confirmatory evidence.

Proxy-package failure does not by itself falsify the core theory, whereas theory-level falsification requires repeated preregistered violations of the terminal-neutrality or variance-compression predictions under validated measurement. The Law of Fairness is a vulnerable and testable scientific hypothesis rather than a metaphysical safeguard. The strongest public scientific targets are preterminal variance compression, terminal concentration near zero, and equivalence-band terminal neutrality under preregistered falsification protocols. Out-of-sample predictive accuracy is used only as the comparative testing criterion between LoF and simpler baseline models. The theoretical framework is explicit; the next step is falsification.

References

- Ames, A. D., Xu, X., Grizzle, J. W., & Tabuada, P. (2017). Control barrier function based quadratic programs for safety critical systems. *IEEE Transactions on Automatic Control*, 62(8), 3861–3876.
- Aubin, J.-P. (2009). *Viability Theory. Systems & Control: Foundations & Applications*. Birkhäuser.
- Bäckman, L., Lindenberger, U., Li, S. C., & Nyberg, L. (2010). Linking cognitive aging to alterations in dopamine neurotransmitter functioning: recent data and future avenues. *Neuroscience & Biobehavioral Reviews*, 34(4), 670–677.
- Baumeister, R. F., Bratslavsky, E., Finkenauer, C., & Vohs, K. D. (2001). Bad is stronger than good. *Review of General Psychology*, 5(4), 323–370.
- Biswas, A., & Borkar, V. S. (2023). Ergodic risk-sensitive control—A survey. *Annual Reviews in Control*, 55, 118–141.
- Bobba-Alves, N., Juster, R. P., & Picard, M. (2022). The energetic cost of allostasis and allostatic load. *Psychoneuroendocrinology*, 146, 105951.
- Cartwright, R., Luten, A., Young, M., Mercer, P., & Bears, M. (1998). Role of REM sleep and dream affect in overnight mood regulation: a study of normal volunteers. *Psychiatry Research*, 81(1), 1–8.
- Casali, A. G., Gosseries, O., Rosanova, M., Boly, M., Sarasso, S., Casali, K. R., Casarotto, S., Bruno, M.-A., Laureys, S., Tononi, G., & Massimini, M. (2013). A theoretically based index of consciousness independent of sensory processing and behavior. *Science Translational Medicine*, 5(198), 198ra105.
- Chaves, P. H. M., Varadhan, R., Lipsitz, L. A., Stein, P. K., Windham, B. G., Tian, J., Fleisher, L. A., Guralnik, J. M., & Fried, L. P. (2008). Physiological complexity underlying heart rate dynamics and frailty status in community-dwelling older women. *Journal of the American Geriatrics Society*, 56(9), 1698–1703.
- Comolatti, R., Pigorini, A., Casarotto, S., Fecchio, M., Faria, G., Sarasso, S., Rosanova, M., Gosseries, O., Boly, M., Bodart, O., Ledoux, D., Brichant, J.-F., Nobili, L., Laureys, S., Tononi, G., Massimini, M., & Casali, A. G. (2019). A fast and general method to empirically estimate the complexity of brain responses to transcranial and intracranial stimulations. *Brain Stimulation*, 12(5), 1280–1289.
- Garcia, J., & Fernández, F. (2015). A comprehensive survey on safe reinforcement learning. *Journal of Machine Learning Research*, 16(1), 1437–1480.
- Germain, A., Buysse, D. J., & Nofzinger, E. (2008). Sleep-specific mechanisms underlying posttraumatic stress disorder: integrative review and neurobiological hypotheses. *Sleep Medicine Reviews*, 12(3), 185–195.
- Gottman, J. M. (1994). *What Predicts Divorce? The Relationship Between Marital Processes and Marital Outcomes*. Lawrence Erlbaum Associates.
- Gottman, J. M. (1999). *The Marriage Clinic: A Scientifically Based Marital Therapy*. W.W. Norton & Company.

- Hesp, C., Smith, R., Parr, T., Allen, M., Friston, K. J., & Ramstead, M. J. D.** (2021). Deeply felt affect: The emergence of valence in deep active inference. *Neural Computation*, 33(2), 398–446.
- Hobson, J. A.** (2009). REM sleep and dreaming: towards a theory of protoconsciousness. *Nature Reviews Neuroscience*, 10(11), 803–813.
- Hofstede, G., Hofstede, G. J., & Minkov, M.** (2010). *Cultures and Organizations: Software of the Mind* (3rd ed.). McGraw-Hill.
- Horvath, S.** (2013). DNA methylation age of human tissues and cell types. *Genome Biology*, 14(10), R115.
- Juster, R. P., McEwen, B. S., & Lupien, S. J.** (2010). Allostatic load biomarkers of chronic stress and impact on health and cognition. *Neuroscience & Biobehavioral Reviews*, 35(1), 2–16.
- Kahneman, D., & Tversky, A.** (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263–291.
- Kara, A. D., & Yüksel, S.** (2024). Partially Observed Optimal Stochastic Control: Regularity, Optimality, Approximations, and Learning. *Proceedings of the 2024 IEEE 63rd Conference on Decision and Control*.
- Kraskov, A., Stögbauer, H., & Grassberger, P.** (2004). Estimating mutual information. *Physical Review E*, 69(6), 066138.
- Li, S. C., & Sikström, S.** (2002). Integrative neurocomputational perspectives on cognitive aging, neuromodulation, and representation. *Neuroscience & Biobehavioral Reviews*, 26(7), 795–808.
- Lipsitz, L. A., & Goldberger, A. L.** (1992). Loss of 'complexity' and aging. Potential applications of fractals and chaos theory to senescence. *JAMA*, 267(13), 1806–1809.
- McEwen, B. S., & Wingfield, J. C.** (2010). What is in a name? Integrating homeostasis, allostasis and stress. *Hormones and Behavior*, 57(2), 105–111.
- Merton, R. C.** (1976). Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3(1–2), 125–144.
- Shenhav, A., Botvinick, M. M., & Cohen, J. D.** (2013). The expected value of control: an integrative theory of anterior cingulate cortex function. *Neuron*, 79(2), 217–240.
- Smith, R., Badcock, P., & Friston, K. J.** (2021). Recent advances in the application of predictive coding and active inference models within clinical neuroscience. *Psychiatry and Clinical Neurosciences*, 75(1), 3–13.
- Vaupel, J. W.** (2010). Biodemography of human ageing. *Nature*, 464(7288), 536–542.
- Wang, Y., & Xu, X.** (2022). Observer-based Control Barrier Functions for Safety Critical Systems. *Proceedings of the 2022 American Control Conference (ACC)*, 709–714.

Appendix A: Candidate Integrated Architecture Beyond the Core Formal Model

This appendix is explicitly speculative and non-core. It does not modify the formal falsifiers in the main document and should not be treated as part of the core falsifiable structure. Its purpose is to record plausible mechanism-level extensions and implementation hypotheses that are compatible with, but not required by, the main formal model.

A. Minimal Hidden-State Exit-Time Control Architecture

The strongest current candidate architecture treats the organism as a partially observed exit-time stochastic controller acting on filtered latent-state beliefs rather than on directly observed ledger states. The minimal latent state is:

$$x(t) = (Y(t), U(t), m(t)),$$

where $Y(t)$ is reserve-normalized latent imbalance, $U(t)$ is biological reserve, and $m(t)$ is a low-dimensional control-mode / channel-feasibility variable. The ledger is then derived rather than primitive:

$$L(t) = U(t)Y(t).$$

Observations are noisy:

$$y(t) = h(x(t)) + n(t),$$

with the stopping proxy $\theta(t)$ treated as an observed auxiliary channel entering the stopping rule rather than as an additional coordinate of the minimal latent state. The controller therefore acts on the filtered belief state:

$$\pi(t) = P(x(t) | \mathcal{Y}_{\{0:t\}}, \theta_{\{0:t\}}),$$

with $\pi(t)$ serving as a sufficient-statistic summary of the joint observation history under the usual filtering assumptions. This is the cleanest currently available architecture for a hidden-state Queue System acting through noisy physiology.

B. Reserve-Shaped Reference Law

The strongest object-level baseline remains the reserve-coupled multiplicative geometry already used in the core model. The reserve process follows:

$$dU(t) = -v(U(t), m(t), t) dt$$

with $U(t)$ treated as strictly positive on the preterminal interval and $v(t) \geq \varepsilon > 0$ in the terminal regime. The normalized latent imbalance follows a reference diffusion:

$$dY(t) = \sigma dW(t),$$

and the ledger is recovered by

$$L(t) = U(t)Y(t).$$

This gives the induced reserve-coupled ledger dynamics:

$$dL(t) = -\left(\frac{v(t)}{U(t)}\right) L(t) dt + \sigma U(t) dW(t)$$

The strongest physical signatures of the Law of Fairness remain drift dominance as $v(t) / U(t)$ increases, variance compression as $\sigma U(t)$ collapses, and terminal concentration of the ledger near zero.

C. Single Steering Layer and Path-Space Cost

The full model should use one primary control mechanism, not multiple overlapping ones. The cleanest candidate is to steer the normalized latent process:

$$dY(t) = u(t) dt + \sigma dW(t),$$

with $u(t) = \mu(\pi(t), t)$, so that the controlled ledger dynamics remain

$$dL(t) = -\left(\frac{v(t)}{U(t)}\right) L(t) dt + U(t) u(t) dt + \sigma U(t) dW(t)$$

This preserves the defining multiplicative identity $L(t) = U(t)Y(t)$ exactly. The Girsanov / KL perspective is then retained only as the path-space interpretation of the same steering layer: under standard controlled-diffusion assumptions and the usual integrability conditions for measure change, a controller deforms a reserve-shaped reference law P into a controlled law Q . In entropy-regularized control formulations, the corresponding steering burden can then be quantified by a relative-entropy cost such as $KL(Q || P)$. This provides the cleanest candidate interpretation of the control burden associated with $\lambda(t)$ as the marginal cost of maintaining admissible trajectories near closure.

D. Exit-Time Horizon and Canonical Costate

The stopping rule remains

$$T = \inf_{t \geq 0} \{ t: \theta(t) \leq \theta_0 \}$$

which preserves the stopping-time firewall because the endpoint remains defined separately from, and not algebraically reducible to, the ledger estimator. The controller-relevant horizon is filtered:

$$H(t) = E[(T - t)^+ | \mathcal{Y}_t]$$

or its preregistered state-space approximation. The terminal-pressure layer is then represented by one constrained exit-time control problem with one canonical costate / marginal value-gradient object $\lambda(t)$. Conditional on a separately specified cost functional, HJB and FBSDE language are best treated as two representations of the same terminal-control structure rather than as separate causal mechanisms.

A useful local approximation remains

$$F_{req} \approx -E[L(t) | \mathcal{Y}_t] / H(t)$$

but only as a small-noise or near-linear heuristic.

E. Derived Admissibility and Belief-Space Pruning

The strongest form of admissibility is not a naked envelope but a derived feasibility condition. If required compensation scales like $|L(t)| / H(t)$, while feasible compensation is bounded by $F_{max}(U(t), m(t))$, then closure feasibility implies:

$$|L(t)| \leq H(t) F_{max}(U(t), m(t)).$$

The reduced-form viability envelope $|L(t)| \leq c U_t^\alpha$ is then retained only as an empirical summary of that deeper structure. Under partial observation, admissibility is naturally belief-based:

$$\pi(t)[|L(t)| \leq c U(t)^\alpha] \geq 1 - \varepsilon.$$

In this framework, the Queue System becomes an admissible-set pruning controller acting on filtered beliefs rather than on ideal latent-state access.

F. Biological Implementation Hierarchy

The most plausible real-world implementation is hierarchical rather than single-channel.

Primary:

- Online cortical valuation / inhibitory control
- Autonomic and endocrine regulation in parallel

Secondary:

- REM / offline affective rebalancing

Terminal physical layer:

- Passive reserve-coupled drift dominance and variance collapse, where the control-mode variable $m(t)$ represents which channels are available, saturated, or degraded; this

architecture makes the model more realistic than a flat list of compensatory channels and provides a principled place for blocked-channel or fallback-channel dynamics.

G. Phenomenological Bridge

The strongest defensible phenomenological bridge is narrow and operational. $F(t)$ is the latent lived valence-rate stream corresponding to reserve-weighted regulatory mismatch and closure-feasibility pressure experienced by the organism. In the phenomenological representation, the unity-gated time integral of that stream is the phenomenological representation of the ledger:

$$L(t) = \int_0^t g_\theta(s) F(s) ds$$

while the stochastic formalization continues to model the same underlying ledger through the reserve-coupled process $L(t) = U(t)Y(t)$. This does not claim a reduction of consciousness to control theory. It claims only that the lived valence stream is the phenomenological correlate of the organism's burden-regulation trajectory under the boundary condition being tested.

H. Exact Closure Status

Exact pathwise closure remains plausible as a latent structural idealization rather than as a theorem of the full hybrid architecture. The strongest current public scientific targets remain preterminal variance compression, terminal concentration near zero, and equivalence-band terminal neutrality within a preregistered empirical framework. The strongest latent strong-form reading is: if the stopping manifold aligns with reserve collapse so that $U(T) = 0$ in the latent process and $Y(T)$ remains finite almost surely, then $L(T) = U(T)Y(T) = 0$ follows almost surely. That strong-form statement remains a structural hypothesis, not yet a proved theorem of the hybrid stack.

I. Candidate Future Refinements Kept Outside the Core

The following ideas remain worth preserving for future work:

- Schrödinger-bridge / entropy-regularized steering as an appendix-level lens on controlled path steering
- Freidlin–Wentzell small-noise asymptotics as an appendix-level sharpening of terminal concentration
- Barrier-slack or normalized-state supermartingale stability theory
- Mean-field social coupling as a future multi-agent extension
- Interoceptive active-inference interpretations as optional phenomenological overlays
- Terminal gamma synchrony as a possible neurophysiological correlate rather than a load-bearing mechanism

These refinements are preserved as non-core extensions while the main formal model remains focused on the strongest current falsifiable structure.